

YERI PARK

Embedded ML Systems Engineer

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Embedded ML systems engineer with four years of experience deploying and optimizing deep neural networks on autonomous-driving embedded platforms – quantization, pruning, and RTOS-based real-time inference on edge SoCs. Currently pursuing an M.S. in Computer Science at Georgia Tech.

EXPERIENCE

Hyundai Motor Company

Researcher, Department of Autonomous Driving SW Development

Seoul, Republic of Korea

Jan 2022 – Present

- Deployed mixed-precision quantization (PTQ/QAT), knowledge distillation, and structured pruning for production autonomous-driving DNNs, reducing inference latency and memory footprint on NVIDIA Orin and Qualcomm SoCs; built the validation framework confirming recovered full-precision accuracy
- Optimized VLM/VLA inference across GPU server clusters and edge devices via automated layer-wise quantization and host<->GPU/NPU memory allocation, cutting end-to-end inference overhead
- Architected an RTOS-based AI inference framework for autonomous embedded systems, integrating car-network interfaces (LIN, CAN, Ethernet, LVDS), multimodal camera/LiDAR/Radar sensor logging, and HILS/SILS validation across NVIDIA, Qualcomm, and TI platforms
- Built a DNN optimization automation framework for GPU-cluster host<->device memory optimization, inference acceleration, and validation testing against original DNN artifacts

Korea Institute of Science and Technology

Undergraduate Research Intern, Robot Vision Lab

Seoul, Republic of Korea

Jun 2021 – Sep 2021

- Designed RGB-D CNN object detection and freespace perception for wearable gait-assist robots on NVIDIA Xavier

Hyundai Motor Company

Undergraduate Research Intern, LiDAR Perception Development

Seoul, Republic of Korea

Sep 2021 – Dec 2021

- Built LiDAR preprocessing, ground segmentation, and point-cloud clustering pipelines; implemented a real-time PCD visualization tool

Ewha Womans University

Undergraduate Research Intern, Digital Signal Processing Lab

Seoul, Republic of Korea

Jun 2019 – Aug 2019

- Developed a human respiration classification model using IR-UWB radar signals

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Online

Sep 2025 – Present

Ewha Womans University

Bachelor of Science in Electronic and Electrical Engineering, GPA: 97.9/100

Seoul, Republic of Korea

Mar 2017 – Feb 2022

SKILLS

Programming: C/C++, Python (PyTorch), Rust, CUDA/OpenCL, Haskell

Embedded & RTOS: FreeRTOS, Linux Kernel, Yocto Project, HILS/SILS Validation

Automotive Protocols: CAN, LIN, Automotive Ethernet, LVDS

ML Systems: TensorRT, ONNX-MLIR, PTQ/QAT, Knowledge Distillation, Pruning, Mixed-Precision Quantization, KV Cache Optimization

Platforms: NVIDIA Orin, Qualcomm SoCs, NVIDIA Xavier, NVIDIA A100, TI

Certifications: ISTQB CTFL, ISTQB AI Testing, NVIDIA CUDA Programming

Languages: Korean (Native), English (Fluent)

ADDITIONAL PROJECTS

Edge AI Network Pruning (Aug 2025) — Applied structured/unstructured pruning to CV networks on NVIDIA Orin. C/C++, TensorRT-LLM, Swin Transformer

Rust-Based ONNX-MLIR Optimization (Apr 2025) — Built a network optimization framework for universal hardware compilation. Rust, Python, C++, ONNX-MLIR

Autonomous Driving LiDAR Perception (Dec 2023) — Ground segmentation, curb detection, free-space estimation, 3D object detection. C/C++, CUDA, OpenCL, PyTorch

HONORS & AWARDS

2nd Prize, Ewha Engineering Capstone Design Contest (2020) — AI-based audience response analysis system

Excellence Award, Korea Electric Power Engineering Camp (2020) — Top 3, national engineering camp

Academic Excellence Scholarship, Ewha Womans University — 4 consecutive semesters (2017–2021)